Domain Attribute:

1) Basic Type
2) Additional clause

Additional Constraints
Rules.

Relational Instance $r(R)$ → dynamic

Apriori Property

Name DOB CGPA

K5, K7 → Candidate Key

ID	NAME
101	a
9	b
1	c
2	d
102	e
4	f

sort

1
2
4
9
101
102

Must be

(I) Informative
(II) Efficient.

ID	
1	3
2	4
4	5
9	2
101	1

← RP

DESIGNNN
Manager

(E1)
MAN-001
MAN-002
SMAN-001

problem → promotion
so SMAN-002
↓
senior.

Lec-4

12.09.21

index → reduced search time significantly

primary key: contains ^Info useful for user.

Foreign Key → can be null & repeated.
→ connect 2 tables.

↳ intuition → object oriented programming (separate objects)

primary key

1) Must be unique

2) can't be null.

recursive SQL / SQL nesting.

→ $R_1 \bigcirc R_2 \Rightarrow R_3$ (another relation)

6 basic operators:

1) select

$\sigma_p(r)$ — r: name of the relation; p: predicate condition

$\sigma_{dept=physics}(instructor)$

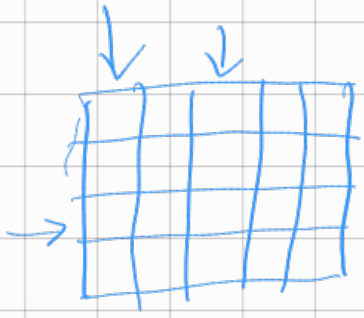
The result of any operation is a relation.

$\sigma_{name=Einstein}\left(\sigma_{dept=physics}(instructor)\right)$

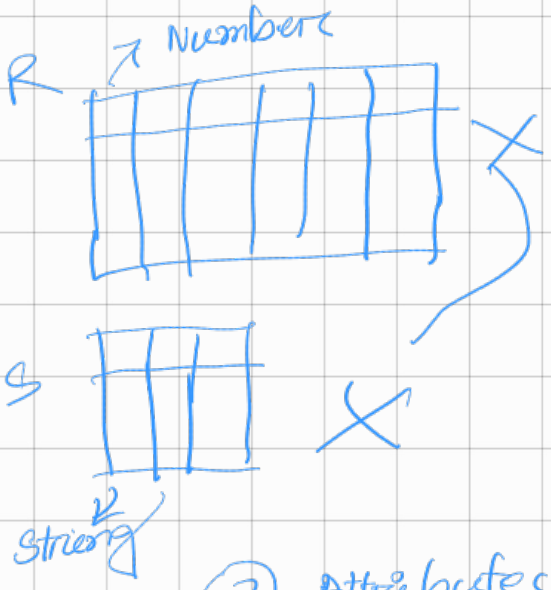
↓

projection.

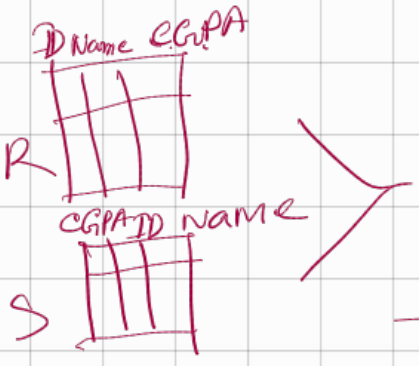
any result of any expression → Relation



for union → ① Must have same number of Attributes.

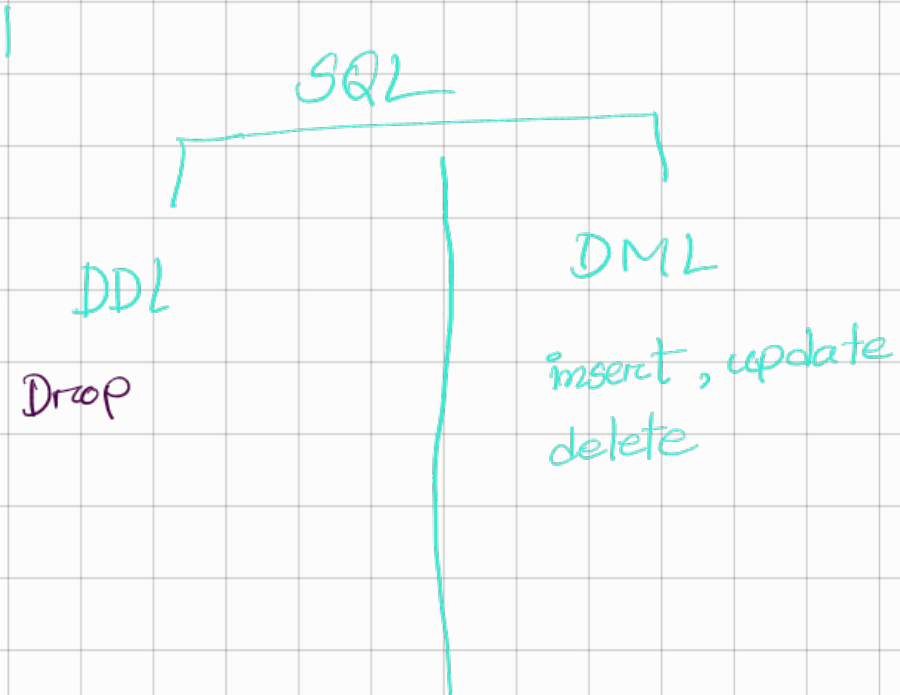


② Attributes must be compatible.



2 condⁿ met.
still disasters
→ combine → ID will be CGPA.

1st Quiz → Next quiz – upto ch 3.



select

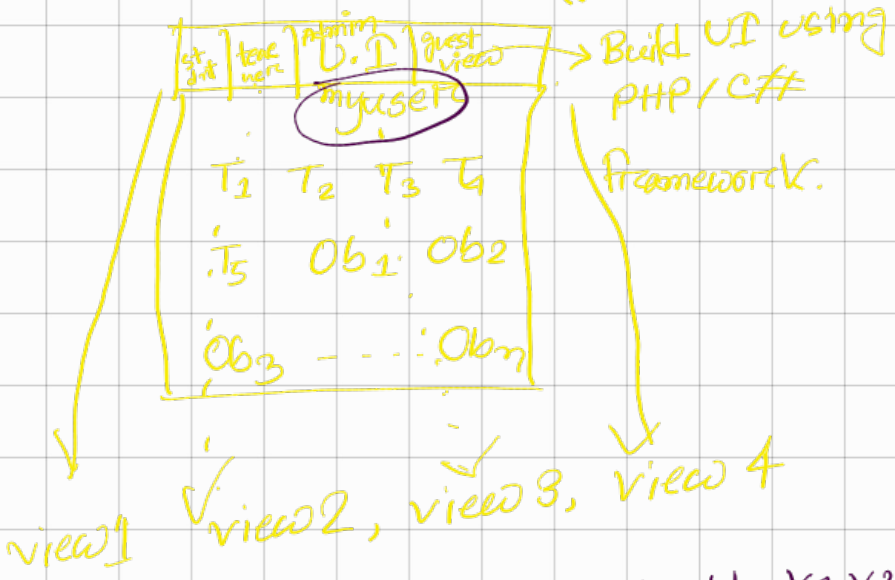
schema : Table.

↳ Oracle

grant DBA to myuser

↓

bundle of privileges.



instead of myuser built V1, V2 ...

varchar(2) → 20 byte/character same in Eng diff in chinese.

Name varchar2(20) → max length of character.

1 chinese character →

Name varchar2(20 char); ⟹ Exam.

x number (1, 1),
(3, 2)
(12, 10)

Domain name Attribute → basic data type

3 to 5 to 4.7.

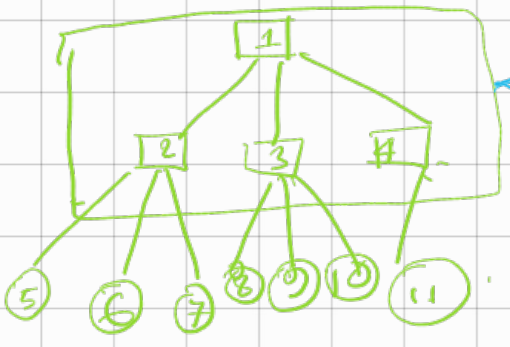
ID : id → some.

Name	Dept	CGPA
A	CSE	3.7 (3.9)
A	EE	3.9
C	CSE	3.5

Select Name, CGPA From S; → whole should be independent.

~~A 3.7 (3.0)~~

A	3.9
C	3.5



Store only this part. Find the bosses.

Linked list → dynamic, self-referencing

EMP
- ID . P.K (primary Key underline)
- Name
- Design
- EBID (Foreign Key) self reference that's why pubicly allowede to give null

Aggregat Function

Where $A.a = B.b$ * write this way in exam

select count(salary)
from E → 1 ? Null
not 0

N	salary
a	100
b	10
c	200



tuples/records finding

select count(ID)
from E;
select count(*)
from E;

group by operator

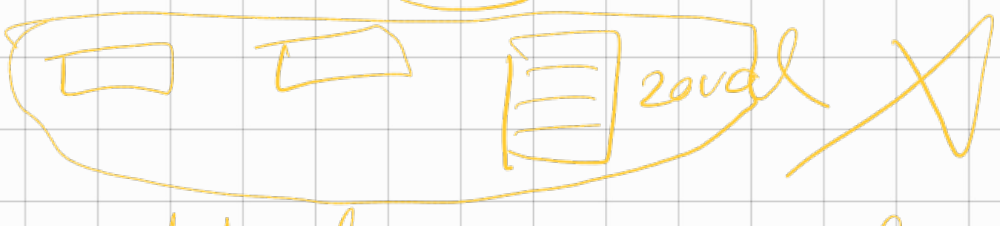
select dept_name and avg(salary)

Depts
- Name => 20 val
- loc
- Budget =>

Dept Budget EES

Instructors
- ID
- Name
- DN
- salary → 1val

select



To reduce all repeated value by min/max func

same values

obtain one value.

Using Having clause

Name/dept → stored val can apply where

Dept	Budget	FES
CSE	1.1	201
EPE	1.1	205
MPE	2.1	205

aggregate (*) > 200;

↓ slide 81

Ch - 1-3 quiz next week.